TRANSMISSION ENGINEERING STANDARD

نقل الكهرباء
National Grid SA

NATIONAL GRID SA
AMMENDMENT-01
S&SD

TES-P-119.26

Revision No. 01

CONTROL SCHEME FOR SUBSTATION EQUIPMENT

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ORIGINATED BY	CERTIFIED BY	APPROVED BY
Engr. Ali M. Mashhour Division Manager, Protection & Substation Automation Specifications	**Engr. Majed A. Al-Suhaibani** Department Manager, Standards & Specifications	**Engr. Abdulaziz A. Al-Motawa** Vice President (A), Engineering
Date of Approval: July 05, 2021		

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Modified Clause #2 and Sub-Clauses shall be read as below:

2.1 The **AVC** panel shall be fabricated in accordance with 31-TMSS-01as applicable for control panels. The devices to be provided are listed in 31-TMSS-01 and alarms, **indications, commands & measurement**s required to be generated from the panel are listed in the **latest revisions of** 38-TMSS-05 **& TES-P-107.02**.

2.2 Automatic Voltage **Controller** (AV**C**) Relay

2.2.1 The AV**C** Relay shall be equipped with integral line…

2.2.2 The AV**C** Relay shall have the facility for testing and obtaining in service reading without disturbing other devices. All auxiliary equipment for raise, lower, blocking, SCADA/**SAS** control and for target and alarm purposes shall be provided. The PT secondary voltage shall be connected to the relay via test facilities.

2.2.3 The AV**C** Relay shall have a digital indications to monitor the tap positions during raise and lower commands. LEDs shall be provided for different status of indications. AV**C** devices shall have LCD screen for measurements, fault and event display and immune to electromagnetic interference.

2.2.4 If the tap position is not in step with other transformers, then 'out of step' **alarm** shall be provided. **Out of step alarm shall not block the operation of OLTC through AVC relay for enough period of time to allow adjustments of taps.**

2,2,5 PC communication port shall be provided on front of AV**C** relay working side.

2.2.6 AV**C** relay shall be in compliance with IEC 61850 **and shall be integrated with SAS**

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2.3 Operating Voltage for **AVC**

The control scheme for **AVC** shall be **fed from 125V DC with separate MCB's at DCDB and AVC panel for each circuit and both shall provide with DC supervision at the end of the loop with alarm signals from MCB trip/off and supervision operated shall be wired to SAS.** The tap changer cabinet local/off/remote selector switch **indications shall be wired to SAS and AVC relay.**

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2.4 The parallel operating scheme

a) The parallel operating scheme shall be equipped with **Circulating Current function** capable of detecting out of step conditions.

b) Master/Follower is only for EHV (unless specified in project technical specifications).

The parallel operating scheme **with Master/Follower** shall be equipped with a blocking scheme capable of detecting out of step conditions no transformer **AVC** shall be capable of being selected to master mode if the transformer is not operating in parallel with one of the other transformers. Selection of master mode on one transformer **AVC** shall automatically force follower mode on the **AVC** of the transformer operating in parallel. Principles based on **Master/Follower** Scheme shall be used for **AVC** functions if specified in the Project Technical Specifications.

"Remote Tap Changer Control (RTCC)" shall be corrected as "**Automatic Voltage Control (AVC)**" everywhere if any.

2.6 Operation Modes.

The **AVC** shall be designed to operate in the Master follower scheme or Circulating current Scheme.

a) The parallel operating scheme shall be equipped with Circulating Current function capable of detecting out of step conditions when circulation current exceed the setting limit in the AVC relay and Out of step alarm shall not block the operation of OLTC through AVC relay for enough period of time to allow adjustments of taps. Scheme shall have a bi-stable relay (for all required selectivity as AVC Auto/Manual, Raise /Lower …etc.)

b) If preferred, the conditions for **Master follower** Scheme shall be included in the PTS of the project. In Master follower Scheme, each mode shall have a **bistable** relay **(for all required selectivity as AVC Auto/Manual, Raise**

/Lower …etc.). The **AVC** shall be provided with an Out of Step scheme to alarm the out of step condition for transformers operating in parallel under a Master-Follower scheme**, Out of step alarm shall not block the operation of OLTC through AVC relay for enough period of time to allow adjustments of taps.**

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2.6.2 **(AVC), the same shall be corrected everywhere if any**
Master /follower is not required **for HV Level & Master/Follower is only for EHV (unless specified in project technical specifications).**

2.7.6 For HV ~~Substations~~ transformer, circulating current scheme shall be followed. Circulating Current Scheme shall be **used to control the tap positions through AVC relay.**
Master /follower scheme is not required for HV Level & Master/Follower is only for EHV (unless specified in project technical specifications)

Add the following

2.7.7 Command for Raise/Lower in Auto mode shall be only from AVC relay.

2.7.8 Auto/ Manual selectivity and indications for AVC Panel shall be executed by push button with Bi-Stable Relay.

2.7.9 In AVC Local position, the Auto/Manual or AVC (raise /lower) commands shall be allowed locally only.

2.7.10 Alarms, indications, commands and measurements from AVC panel shall be as per 38-TMSS-05 and TES-P-107.02.

2.7.11 AVC operational mode shall be as indicted below in table 1,

Table 1

Authority to control					Remarks
AVC Control mode	**AVC Selection mode**	**Local command**	**Remote command**	**AVC command**	
REMOTE	**AUTO**	**NO**	**NO**	**YES**	1- Local/Remote selectivity can be selected only locally 2- Auto / Manual selectivity can be selected only remotely
REMOTE	**MANUAL**	**NO**	**YES**	**NO**	1- Local/Remote selectivity can be selected only locally 2- Auto / Manual selectivity can be selected only remotely
LOCAL	**AUTO**	**NO**	**NO**	**YES**	1- Local/Remote selectivity can be selected only locally 2- Auto/Manual selectivity can be selected only locally.
LOCAL	**MANUAL**	**YES**	**NO**	**NO**	1- Local/Remote selectivity can be selected only locally 2- The commands in this condition shall not be executed through AVC relay,

					and it should have hardwire (Through push button and bi stable relay). 3- It can be used even if AVC relay is faulty.

2.8 SCADA/**SAS** Control of **AVC.**

2.8.1 If either the **AVC** Local/ mote switch is selected to "LOCAL" or "OFF", no control from SCADA/**SAS** ~~is available~~ **is authorized. And alarm shall be send to SAS and control center, and the selectivity of Auto/Manual (AV~~R~~ C In/Out) selection and control for Raise /Lower command can't be achieved through SCADA/ SAS if AVC panel is in Local position even if the OLTC LCC selector switch is set to "REMOTE".**

2.8.2 If **AVC** Local/**Remote** selector switch is selected to **Remote** and the OLTC LCC selector switch is set to "REMOTE", then control of the **AVC** cubicle shall be accessed through SCADA/ **SAS**.

2.8.3 Control is achieved by selecting the Auto/Manual **(**AV~~R~~ **C** In/Out) selection to "MANUAL". (AV**C** Out), the **automatic** raise and lower functions of the AV**C** are blocked and the red indicating LED labeled "MANUAL" is illuminated **but the Control for Raise /Lower can be achieved Manually.** When selected to "AUTO" (AVC In), the a**utomatic** raise and lower function shall be through AVC and LED labeled "AUTO" is illuminated.

2.8.4 SCADA/**SAS** shall monitor the status and control the mode of operation by energizing **bi-stable** relays **if the AVC Local/Remote selector switch is selected to "Remote while if the selectivity is** in local mode, these **bi-stable** relays are controlled **locally** by the scheme push buttons. SCADA/**SAS** shall monitor all the taps in the tap changer.

2.8.5 SCADA/**SAS** shall be able to select Master or Independent and Auto or Manual mode.

2.8.6 Once SCADA/**SAS** has selected to "MANUAL" (AV~~R~~**C** Out), the tap position can be manually raised or lowered in Master or Independent modes. Tap changer operation shall be in such a way that a SCADA/**SAS** raise command corresponds to a raise in the tap position and a raise in the transformer low side voltage.

2.8.7 SCADA/**SAS** requirements shall be as per **latest version of** TES-P-119.27 **and TES-P-107.02**

Add the following,

2.8.8 The automatic (AUTO) control mode shall not be affected by AVC selection mode status; it should remain control the bus voltage automatically in both selection (remote & local).

2.9.4 OLTC can be controlled manually from push buttons from AVC panel even If AVC relay is faulty.

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2.10.2 If applicable for a three transformer station, Automatic Bus Transfer Scheme (ABTS) control shall be possible as long as Local/Off/Remote switch in the OLTC LCC is in the REMOTE position regardless of the position of the RTCC Local/Supervisory switch or pre-selected modes. These control inputs shall enable the ABTS to block or enable automatic operation, match taps prior to paralleling transformers and select a new Master RTCC as required. **For HV Substations, ACSE shall not block its operation even if taps are not matched and taps shall be matched via AVC relay accordingly**.

Modified Clause #3 & Sub-Clauses shall be read as below:

3.1 Intelligent Electronic Device (IED).

3.1.1 The Automatic **Control Switching Equipment (ACSE)** shall be accomplished by the use of Microprocessor based **IED** designed for use in the substation environment. It shall have the watchdog system capability to monitor the **IED** circuits and identify the defective circuit and give alarms. These alarms shall be monitored by SCADA/**SAS**. It shall have an on-off switch which shall also be operated and monitored by SCADA/**SAS**. The programming procedures shall be easily done without special knowledge of programming. The Human Machine Interface programming equipment shall be provided. The **IED** shall meet the **latest revision of** IEC 61131 and/or equivalent ANSI/IEEE.

3.1.2 The **IED** shall be flexible, extendible if any input/output is needed. The **IED** program shall be able to be read-in and modified by SEC, if needed in future. The equipment needed to reprogram the microprocessor such as servicing case, programming unit and video display unit shall be included in the supply in order to accomplish this task at the field.

3.1.3 The **IED** shall be equipped with its own DC/DC converter conforming to the requirements of IEC standard.

3.1.4 The **IED** shall be completely devoted to **ACSE** and shall not be used as an annunciator.

3.1.5 Under SAS, the **ACSE** panel shall be integrated and provided with dedicated control IED to carryout **ACSE** functions.

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ABTS shall be corrected as ACSE everywhere if any.

3.3 ACSE/TOP shall control all the five circuit breakers (Incomers and Bus Section) according to all possible tripping scenarios.

3.3.1 ACSE Auto Operation mode Requirements:

a. ACSE Selection should be on IN position.
b. All transformers GT1, GT2 and GT3 should be in service/energized from high voltage side.
c. ACSE shall control the five circuit breakers in distribution voltage level, where the four CBs are closed/in service and the fifth one is a hot standby breaker.
d. ACSE scheme operates, if 4 out of 5 controlled CBs are closed/in service, the VT of the fifth CB hot standby is healthy, ACSE IN and ACSE ready.
e. ACSE Auto Mode Operation should not concern by the control panel Local /Remote selectivity.
f. ACSE Auto Operation should check the status of the AVC mode (in Auto) and AVC relay healthy otherwise ACSE operation shall be blocked.
g. ACSE Operation should not be blocked by the AVC panel in case of Out of Step Alarms.
h. Capacitor CB status is required in the ACSE logics.
i. ACSE should be initiated by the following permanent protection signals, 87T, LV REF, HV REF, HV CBF, HV BB, LV CBF, LV BB (of the concerned bus) and Mechanical protection except (LV 49W and HV 49W).
j. Fire Protection Signal should block the ACSE operation.
k. ACSE initiation for Breaker fail protection and bus bar protection operation (if it is not related to the faulty bus-bar).
l. ACSE should be blocked by the following temporary protection signals, LV 51/51N, HV 51/51N, SBEF and Mechanical protection LV 49W and HV 49W.

3.3.2 TOP Manual Operation Mode Requirements:

a. TOP Selection should be on IN position.
b. TOP scheme can be operated locally or remotely.
c. All transformers GT1, GT 2 and GT3 should be in service/energized for high voltage side.
d. TOP shall control the five circuit breakers in distribution voltage level, where four CBs are closed/in service and the fifth one is hot standby.
e. TOP scheme operates, if 4 out of 5 controlled CBs are closed/in service, TOP Ready and TOP IN.
f. Operator (Local) and Control Center (Remote) must have the facility to close the fifth breaker after TOP selection from ACSE panel or SAS/HMI or control center.
g. Operator (Local) and Control Center (Remote) should have the facility to select any breaker to open immediately after closing hot-standby circuit breaker.
h. The selected breaker should not be open if the standby circuit breaker does not close.
i. If the selected CB does not open after closing the hot standby circuit breaker, the hot standby CB should be open after a short time delay (30 sec).
j. TOP Selection time shall be 2 min and it shall be re-set automatically after this time.
k. During TOP Selection time, if the operator/Control Center select other CB to open, the first selection shall be re-set automatically.

3.3.3 ACSE/TOP NOT READY Scheme:

ACSE / TOP should issue Not Ready signal if,

a. **More than one circuit breaker of the five-controlled CB's are open in normal condition.**
b. **One of the five-controlled circuit breaker is not in service.**
c. **If any one of the transformers HV side CB not closed or not in service in the normal condition.**
d. **One of the five-controlled circuit breaker is not in remote mode. All the concerned breakers should be in remote mode for ACSE/TOP Ready.**
e. **ACSE/TOP should be ready for some time even though the condition is not 4 out of 5. Once the stand by CB closed, ACSE/TOP should be Not Ready.**
f. **ACSE Operational Mode Selection shall be as per Table 2,**

Table 2

Authority to control				
ACSE/TOP Selection Mode	**Control Mode**	**Local Command (Manual command)**	**Remote Command (Manual command)**	**Controller Command (Automatic command)**
ACSE IN	**LOCAL**	**NO**	**NO**	**YES**
	REMOTE	**NO**	**NO**	**YES**
ACSE OUT	**LOCAL**	**NO**	**NO**	**NO**
	REMOTE	**NO**	**NO**	**NO**
TOP IN	**LOCAL**	**YES**	**NO**	**NO**
	REMOTE	**NO**	**YES**	**NO**
TOP OUT	**LOCAL**	**NO**	**NO**	**NO**
	REMOTE	**NO**	**NO**	**NO**

Moreover, in case, if TOP is not on IN position, a hardwire interlock to be implemented to prevent the fifth CB from closing.

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Modified Clause #4 and Sub-Clauses shall be read as below:

4.1 AUTOMATIC CONTROL OF CAPACITOR BANK SYSTEM (ACCS) (FOR SUBSTATIONS WITH THREE TRANSFORMERS)

The automatic control of the capacitor banks shall be accomplished by the use of the **IED**. The panel to house **IED** and their associated devices of ACCS shall conform to 31-TMSS-01. The **IED** shall be designed to be used in the substation environment. It shall have the watchdog system capability to monitor the **IED** circuits and identify the defective circuit and give alarms. These alarms shall also be operated and monitored by SCADA/**SAS**. The programming procedures shall have following features.

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4.1.4 shall be modified and read as below,

4. CASE-IV

Capacitor Bank-1 (C-1) shall be switched ON when the total reactive power of Power Transformers T-1, T-2 & T-3 measured (HV side) is above the set value of the reactive power (MVAR-3) and the duration of this reactive power exceeds the set value of time (t-3). If the total reactive power of the Transformers T-1, T-2 & T-3 measured is below the set value of reactive power (MVAR-4) and the duration of reactive power below the set value of reactive power is above the set value of time (T-4), the Capacitor Bank -1 (C-1) shall be switched OFF or remain in OFF condition.

Capacitor Bank -2 (C-2) shall be switched ON when the total reactive power of Power Transformers T-1, T-2 & T-3 measured (HV side) is above the set value of reactive power (MVAR-5) and the duration of this reactive power exceeds the set value of time (t-5). If the total reactive power of Transformers T-1, T-2 & T-3 measured is below the set value of reactive power (MVAR-6) and the duration of reactive power below the set value of reactive power, is above the set value of time (t-6), the Capacitor Bank-2 (C-2) shall be switched OFF or remain in OFF condition**. (in case of T1 & T2 are parallel)**

Capacitor Bank -2 (C-2) shall be switched ON when the total reactive power of Power Transformers T-1, T-2 & T-3 measured (HV side) is above the set value of reactive power (MVAR-3) and the duration of this reactive power exceeds the set value of time (t-3). If the total reactive power of Transformers T-1, T-2 & T-3 measured is below the set value of reactive power (MVAR-4) and the duration of reactive power below the set value of reactive power, is above the set value of time (t-4), the Capacitor Bank-2 (C-2) shall be switched OFF or remain in OFF condition. (in case of T2 & T3 are parallel)

Capacitor Bank-3 (C-3) shall be switched ON when total reactive power of Power Transformers T-1, T-2 & T-3 measured (HV side) is above the set value of reactive power (MVAR-5) and the duration of this reactive power exceeds the set value of time (t-5). If the total reactive power of Transformers T-1, T-2 & T-3 measured is below the set value of reactive power (MVAR-6) and the duration of reactive power below set value of reactive power is above the set value of time (t-6), the Capacitor Bank-3 (C-3) shall be switched OFF or remain in OFF condition.

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4.1.5 Shall be Modified and read as below,

The minimum inputs to the ACCS shall consist of status of all **HV and MV** incomer (Transformer) CBs, bus tie CBs, each 33 kV/13.8 kV incomer voltage, each 33 kV/13.8 kV bus voltage, each 33 kV/13.8 kV Capacitor Bank bus voltage, each Power Transformer's Varreadings on the HV side, and status of lockout relays of incomer CBs/Capacitor CBs. Moreover, provision for overload protection shall be incorporated.

During ACCS Auto Mode, when any 13.8KV incomer CB switched OFF, the associated capacitor bank shall also switched OFF before 13.8KV Bus section CB switched ON. This condition will not be applicable when ACCS is on Manual mode. The CB of the Capacitor bank once switched OFF shall not switch ON within 10 minutes. Auto / Manual and Local / Remote selection shall be provided for the ACCS. In any case the 13.8 kV incomer CBs of Transformers T-1, T-2 & T-3 are switched ON, the associated Capacitor shall not switch ON within 10 minutes. Also two nos. Capacitor banks shall not be connected to a single Transformer.

The ACCS shall prevent undesired switching 'ON' operation of CBs of associated Capacitor banks in any one of the following cases:

i 33 kV/13.8 kV Bus fault (86 B1 or 86 B2 or 86B3) operates
ii 33 kV/13.8 kV Capacitor feeder unclear faults (main feeder CB or dedicated Capacitor feeder CB Lockout relay operates)
iii 33 kV/13.8 kV Bus Tie Lockout operates
iv 33 kV/13.8 kV Incomer Lockout operates

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4.1.5 The following shall be added,

iv. If any bus de-energized for any reason, the associated capacitor must be open/trip.
v. If any capacitor bank CB is not in remote and in service mode in switchgear panel, ACCS in Auto mode should not issue command to this capacitor.
vi. During ACCS manual operation mode, both functions. (Transformer loading measurement MVAR and associated bus voltage measurement) are not required. Authority shall be given to control and field operation through hard wiring or switchgear BCU (not through ACCS relay) to put the capacitor bank in service or not. Unless if there are any safety requirements or protection interlock prevent CB, operation shall be executed by hardware from scheme and software from switchgear BCU.
vii. Auto/Manual operation mode selectivity shall be executed (locally in the panel using push buttons or spring return selector switch) through Bi-stable relays so that manual operation can be carried out from Control Centre. It should not be selected from ACCS Controller.
viii. Manual operation shall not be disturbed if ACCS relay is Faulty/OFF.
ix. Alarms, indications, commands and measurements from ACCS panel shall be as per TES-P-107.02.
x. ACCS ON/OFF Selector Switch shall be provided in the panel.

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4.1.6 The range of setting shall be as per table 3 below,

Table 3

ACCS FINAL SETTINGS FOR 3 TRANSFORMERS SYSTEM											
CASE 1: Bus Section A/B Open and Bus Section B/C Open											
	SET Mvar FOR CLOSING						**SET Mvar FOR OPENING**				
		Mvar		**TIME**				**Mvar**		**TIME**	
QC1	CLOSE	9	Mvar-1	5 min.	t1		OPEN	1	Mvar-2	5 min.	t2
QC2	CLOSE	9	Mvar-1	5 min.	t1		OPEN	1	Mvar-2	5 min.	t2
QC3	CLOSE	9	Mvar-1	5 min.	t1		OPEN	1	Mvar-2	5 min.	t2
	SET VOLTAGE FOR CLOSING						**SET VOLTAGE FOR OPENING**				
		Voltage		**TIME**				**Voltage**		**TIME**	
QC1	CLOSE	12.1	V2	60 sec.	t10		OPEN	14.6	V1	5 sec.	t9
QC2	CLOSE	12.1	V2	60 sec.	t10		OPEN	14.6	V1	5 sec.	t9
QC3	CLOSE	12.1	V2	60 sec.	t10		OPEN	14.6	V1	5 sec.	t9
CASE 2: Bus Section A/B Open and Bus Section B/C Closed											
	SET Mvar FOR CLOSING						**SET Mvar FOR OPENING**				
		Mvar		**TIME**				**Mvar**		**TIME**	
QC1	CLOSE	9	Mvar-1	5 min.	t1		OPEN	1	Mvar-2	5 min.	t2
QC2	CLOSE	10	Mvar-3	5 min.	t3		OPEN	2	Mvar-4	10 min.	t4
QC3	CLOSE	11	Mvar-5	10 min.	t5		OPEN	3	Mvar-6	5 min.	t6
	SET VOLTAGE FOR CLOSING						**SET VOLTAGE FOR OPENING**				
		Voltage		**TIME**				**Voltage**		**TIME**	
QC1	CLOSE	12.1	V2	60 sec.	t10		OPEN	14.6	V1	5 sec.	t9
QC2	CLOSE	12.1	V2	60 sec.	t10		OPEN	14.6	V1	10 sec.	t7
QC3	CLOSE	12.1	V2	120 sec.	t8		OPEN	14.6	V1	5 sec.	t9
CASE 3: Bus Section A/B Closed and Bus Section B/C Open											
	SET Mvar FOR CLOSING						**SET Mvar FOR OPENING**				
		Mvar		TIME				**Mvar**		**TIME**	
QC1	CLOSE	10	Mvar-3	5 min.	t3		OPEN	2	Mvar-4	10 min.	t4
QC2	CLOSE	11	Mvar-5	10 min.	t5		OPEN	3	Mvar-6	5 min.	t6
QC3	CLOSE	9	Mvar-1	5 min.	t1		OPEN	1	Mvar-2	5 min.	t2
		Voltage		**TIME**				**Voltage**		**TIME**	
QC1	CLOSE	12.1	V2	60 sec.	t10		OPEN	14.6	V1	10 sec.	t7
QC2	CLOSE	12.1	V2	120 sec.	t8		OPEN	14.6	V1	5 sec.	t9
QC3	CLOSE	12.1	V2	60 sec.	t10		OPEN	14.6	V1	5 sec.	t9
CASE 4: Bus Section A/B Closed, Bus Section B/C Closed and Incomer z Open											
	SET Mvar FOR CLOSING						**SET Mvar FOR OPENING**				
		Mvar		**TIME**				**Mvar**		**TIME**	
QCx	CLOSE	10	Mvar-3	5 min.	t3		OPEN	2	Mvar-4	10 min.	t4
QCy	CLOSE	11	Mvar-5	10 min.	t5		OPEN	3	Mvar-6	5 min.	t6
		Voltage		**TIME**				**Voltage**		**TIME**	
QCx	CLOSE	12.1	V2	60 sec.	t10		OPEN	14.6	V1	10 sec.	t7
QCy	CLOSE	12.1	V2	120 sec.	t8		OPEN	14.6	V1	5 sec.	t9

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4.2 The following shall be modified,

The automatic control of Capacitor banks shall be accompanied by the use of PLC/**IED**. The panel to house PLC/**IED** and other associated devices of ACCS shall conform to 31-TMSS-01. The PLC/**IED** shall be designed to be used in the substation environment. It shall have the watchdog system capability to monitor the PLC/**IED** circuit and identify the defective circuits and give alarms. These alarms shall be monitored by SCADA/**SAS**. It shall have an ON/OFF switch which will also be monitored by SCADA/**SAS**.

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Saudi Electricity Company

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1	November 29, 2010	Incorporated Substation Automation Scheme (SAS) and Automatic Capacotor Control Scheme (ACCS) in the Standard
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CONTROL SCHEMES

FOR

SUBSTATION EQUIPMENTS

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ORIGINATED BY	CERTIFIED BY	APPROVED BY
Engr. Abdulrahman G. Al-Ghayhab Division Manager (A), Transmission Substation Specifications	**Engr. Abdullah M. Al-Garni** Department Manager, Transmission Standards & Specifications	**Engr. Saleh N. Al-Sohaibani** Executive Director, Engineering & Projects
Date of Approval: November 29, 2010		

TRANSMISSION ENGINEERING STANDARD

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4.2 Automatic Capacitor Control Scheme For Two Transformers

5.0 BIBLIOGRAPHY

1.0 PURPOSE AND SCOPE

The purpose of this chapter is to highlight the requirements for Remote Tap Changer Control (RTCC), Automatic Bus Transfer Scheme (ABTS) and Automatic Capacitor Control Scheme (ACCS) at Medium Voltage in Substations in SEC System.

The requirements for Remote Tap Changer Control (RTCC), Automatic Bus Transfer Scheme (ABTS) and Automatic Capacitor Control Scheme (ACCS) shall be applicable for three-transformer substation and two-transformer substation.

2.0 REMOTE TAP CHANGER CONTROL (RTCC)

2.1 RTCC Panel

The RTCC panel shall be fabricated in accordance with 31-TMSS-01 as applicable for control panels. The devices to be provided are listed in 31-TMSS-01 and alarms required to be generated from the panel are listed in 38-TMSS-05.

2.2 Automatic Voltage Regulating (AVR) Relay

2.2.1 The AVR Relay shall be equipped with integral line drop compensator, inverse and definite time characteristic and reverse reactance or circulating current compensation. Initial and intertap time delays shall be selectable inverse or definite time characteristics. The relay shall have undervoltage blocking and overcurrent blocking features in addition to overvoltage supervision features.

2.2.2 The AVR Relay shall have the facility for testing and obtaining in service reading without disturbing other devices. All auxiliary equipment for raise, lower, blocking, SCADA control and for target and alarm purposes shall be provided. The PT secondary voltage shall be connected to the relay via test facilities.

2.2.3 The AVR Relay shall have a digital indication to monitor the tap positions during raise and lower commands. LEDs shall be provided for different status of indications. AVR devices shall have LCD screen for measurements, fault and event display and immune to electromagnetic interference.

2.2.4 If the tap position is not in step with other Transformers, then ‘out of step’ indication shall be provided.

2.2.5 PC communication port shall be provided on front of AVR relay working side.

2.2.6 AVR Relay shall be in compliance with IEC 61850

2.3 Operating Voltage for RTCC

The control scheme for RTCC shall be 110 V_{ac} operated. The 110 V_{ac} (55V-0-55V) control voltage shall be derived from the tap changer motor supply via a 220/110V_{ac} or 480/110 V_{ac} or 380/110 V_{ac} step down transformer with a center tap grounded secondary located in the power transformer Local Control Cabinet. The RTCC supply shall be via the tap changer cabinet local/off/remote selector switch isolating contacts.

2.4 Parallel Operating Scheme

The parallel operating scheme shall be equipped with a blocking scheme capable of detecting out of step conditions. No transformer RTCC shall be capable of being selected to master mode if the transformer is not operating in parallel with one of the other transformers. Selection of master mode on one transformer RTCC shall automatically force follower mode on the RTCC of the transformer operating in parallel. Principles based on Circulating Current Scheme shall be used for RTCC functions if specified in the Project Technical Specifications.

2.5 On Load Tap Changer Local Control Cabinet (OLTC LCC)

The On Load Tap Changer Local Control Cabinet (OLTC LCC) shall be equipped with a three position Local/Off/Remote selector switch, which is lockable by padlock in the “OFF” position. In the “OFF” position, all supply to the OLTC LCC and RTCC cubicle is isolated **(**except a.c supply to panel heater). In the LOCAL position, control is effective from the OLTC LCC Raise and Lower push buttons. In the REMOTE position, control is effective from the RTCC controls or from SCADA via the Local/Supervisory selector switch installed on the RTCC.

Under SAS, the transformer OLTC shall derive the RTCC functions through IED located in transformer bay control panel and all functions of the transformer shall be integrated with SAS.

2.6 Operation Modes

The RTCC shall be designed to operate in the Master follower scheme or Circulating current Scheme. If preferred, the conditions for Circulating Current Scheme shall be included in the PTS of the project. In Master follower Scheme, each mode shall have a latching relay with electrical reset. The RTCC shall be provided with an Out of Step scheme to alarm the out of step condition for transformers operating in parallel under a Master-Follower scheme.

2.6.1 Master-Automatic Mode

Raise and lower commands from the local AVR Relay are used to effect changes in the locally controlled tap changer and the tap changer of the transformer being operated in parallel.

2.6.2 Master-Manual Mode

Raise and lower commands from the local RTCC and OLTC Local Control Cabinet RAISE and LOWER control push buttons (or SCADA/SAS) are used to effect changes in the locally controlled tap changer and the tap changer of the transformer being operated in parallel.

2.6.3 Follower Mode

Raise and lower commands from the paralleled Master selected transformer RTCC are used to effect changes in the locally controlled tap changer (local RTCC Auto/Manual selection has no effect on operation).

2.6.4 Independent-Automatic Mode

Raise and lower commands from the local AVR Relay are used to effect changes in the locally controlled tap changer.

2.6.5 Independent-Manual Mode

Raise and lower commands from the local RTCC and OLTC Local Control Cabinet RAISE and LOWER control push buttons (or SCADA/SAS) are used to effect changes in the locally controlled tap changer.

2.7 Conditions for Selection of Operation Modes

2.7.1 The Master and Follower operation modes shall not be selectable when the transformer is not actually being operated in parallel with another transformer and RTCCs operating in Master/Follower modes will be automatically reset to Independent mode if the paralleling condition is broken.

2.7.2 Selection of Independent mode on either a Master or Follower mode RTCC shall force the paralleled transformer's RTCCs to Independent mode.

2.7.3 Selection of a Follower mode RTCC to Master shall cause the existing Master RTCC to be forced to Follower mode if the transformers are operating in parallel.

2.7.4 Independent-Automatic mode selection and/or operation shall not be possible if operating in parallel with another transformer.

2.7.5 If transformers operating in parallel are selected to Independent mode, then Manual operating mode shall be forced automatically.

2.7.6 Circulating Current Scheme shall be imposed if the requirements are explicitly defined and included in the Project Technical Specifications.

2.8 SCADA Control of RTCC

2.8.1 If either the RTCC Local/Supervisory selector switch is selected to "LOCAL" or the OLTC LCC Local/Off/Remote switch is selected to "LOCAL" or "OFF", no control from SCADA is available.

2.8.2 If RTCC Local/Supervisory selector switch is selected to "SUPERVISORY" and the OLTC LCC selector switch is set to "REMOTE", then control of the RTCC cubicle shall be accessed through SCADA.

2.8.3 Control is achieved by selecting the Auto/Manual (AVR In/Out) selection relay to "MANUAL". When selected to "MANUAL" (AVR Out), the raise and lower functions of the AVR are blocked and the red indicating LED labeled "MANUAL" is illuminated. When selected to "AUTO" (AVR In), the raise and lower function shall be through AVR and LED labeled "AUTO" is illuminated.

2.8.4 SCADA shall monitor the status and control the mode of operation by energizing latching relays that have electrical reset coils.When in local mode, these latching relays are controlled by the scheme push buttons.SCADA shall monitor all the taps in the tap changer.

2.8.5 SCADA shall be able to select Master or Independent and Auto or Manual mode.

2.8.6 Once SCADA has selected to "MANUAL" (AVR Out), the tap position can be manually raised or lowered in Master or Independent modes. Tap changer operation shall be in such a way that a SCADA raise command corresponds to a raise in the tap position and a raise in the transformer low side voltage.

2.8.7 SCADA requirements shall be as per TES-P-119.27.

2.9 Local Control of RTCC

2.9.1 If either the RTCC Local/Supervisory switch is selected to SUPERVISORY or the OLTC LCC Local/Off/Remote selector switch is selected to LOCAL or OFF, no controls from Local RTCC controls shall be available.

2.9.2 If the RTCC Local/Supervisory switch is selected to LOCAL and OLTC LCC Local/Off/Remote selector switch is selected to REMOTE, then the control of the RTCC cubicle may be accessed through RTCC local controls.

2.9.3 The RTCC local controls shall only be able to select valid operating modes as defined in clause 2.7 above. Local control shall be able to select Auto/Manual, Master or Independent and shall have manual Raise and Lower control after selecting to Master or Independent.

2.10 RTCC for Substation Automation

2.10.1 If applicable for a two transformer station with normally closed bus tie and operating under a master/follower scheme, opening of either transformer MV incomer or bus tie breaker shall automatically select each transformer RTCC to independent auto mode.

2.10.2 If applicable for a three transformer station, Automatic Bus Transfer Scheme (ABTS) control shall be possible as long as Local/Off/Remote switch in the OLTC LCC is in the REMOTE position regardless of the position of the RTCC Local/Supervisory switch or pre-selected modes. These control inputs shall enable the ABTS to block or enable automatic operation, match taps prior to paralleling transformers and select a new Master RTCC as required.

2.10.3 The Substation Automation System (SAS), if used for control of the substation equipments, shall be suitable for remote control from the Power Control Centre through SCADA system and gateway. The bay control panels (BCPs) shall consist of Bay control Units (BCUs)/Bay protection Units (BPUs)/Bay emergency control Units (BECUs) instead of conventional local control panels(LCPs). BCPs shall have control IEDs having local Human Machine Interface (HMI) and provide the functions of Transducers and Interposing relays. Bay control Panels shall be completely wired and adequate terminal blocks shall be provided for termination of external incoming cables.

3 11/13.8/33/34.5 kV AUTOMATIC BUS TRANSFER SCHEME (ABTS)

3.1 Programmable Logic Controller

3.1.1 The Automatic Bus Transfer Scheme shall be accomplished by the use of Microprocessor based Programmable Logic Controller (PLC) designed for use in the substation environment. It shall have the watchdog system capability to monitor the PLC circuits and identify the defective circuit and give alarms. These alarms shall be monitored by SCADA. It shall have an on-off switch which shall also be operated and monitored by SCADA. The programming procedures shall be easily done without special knowledge of programming. The Human Machine Interface programming equipment shall be provided. The PLC shall meet the IEC 61131 and/or equivalent ANSI/IEEE.

3.1.2 The PLC shall be flexible, extendible if any input/output is needed. The PLC program shall be able to be read-in and modified by SEC, if needed in future. The equipment needed to reprogram the microprocessor such as servicing case, programming unit and video display unit shall be included in the supply in order to accomplish this task at the field.

3.1.3 The PLC shall be equipped with its own DC/DC converter conforming to the requirements of IEC standard.

3.1.4 The PLC shall be completely devoted to ABTS and shall not be used as an annunciator.

3.1.5 Under SAS, the ABTS panel shall be integrated and provided with dedicated control IED to carryout ABTS functions.

3.2 Automatic Switching Scheme for Two Transformers

The ABTS scheme operation shall be initiated from Under Voltage condition by a under voltage relay connected on the transformer incomer and the logic shall be as follows:

3.2.1 Tripping of transformer incomer already energized (first stage):

a. ABTS is set on AUTO.

b. ABTS shall check that there is no voltage on the new incomer through the incomer under voltage relay (27) and fuse failure supervision relay (98).

c. ABTS shall check that other bus is healthy and energized through the bus under voltage/over voltage relay (27/59).

d. ABTS shall check that concerned bus is dead through the bus under voltage/over voltage relay (27/59).

e. ABTS shall check that the new incomer is in service and closed.

f. Then ABTS will trip the already energized incomer of the transformer after a time delay.

3.2.2 Tripping of Bus Tie to satisfy the requirement in clause 2.10.1:

a. ABTS shall check the status that first incomer is closed.

b. ABTS shall check the status that second incomer is closed.

c. ABTS shall check the status that bus tie breaker is closed.

d. After a time delay ABTS will trip the bus tie breaker.

3.2.3 Closing of bus tie breaker:

a. ABTS is set on auto.

b. ABTS shall check that there is no voltage on the new incomer through the incomer under voltage relay (27) and fuse failure supervision relay (98).

c. ABTS shall check that other bus is healthy and energized through the bus under voltage/over voltage relay (27/59).

d. ABTS shall check that concerned bus is dead through the bus under voltage/over voltage relay (27/59).

e. ABTS shall check that the new incomer is in service and open.

f. ABTS shall check that other incomer is in service and closed.

g. ABTS shall check that bus tie breaker is in service.

h. ABTS shall check that No operation of 86-2 (reset).

i. ABTS shall check that No operation of 86B1 (reset).

j. ABTS shall check that No operation of 86B2 (reset).

k. ABTS shall check that No operation of 86BT (reset).

l. After a time delay ABTS will close the bus tie breaker.

3.3 Automatic Switching Scheme for Three Transformers

During normal operation, two (2) transformers shall run in parallel and the third transformer shall operate independently. In the event of a transformer failure, the remaining two (2) transformers shall operate in parallel to feed the entire substation load (on all three bus-sections) without interruption of supply.

Minimum requirements of the scheme are:

3.3.1. The arrangement of power transformer lockout relays, as applicable, shall be as follows:

Lockout 86-1 shall be initiated by 87T and 87REF.
Lockout 86-2 shall be initiated by 51GLV and 51HV.
Lockout 86-3 shall be initiated by 50HV, 50/51NHV, 67/67N and mechanical protections.

3.3.2 The ABTS shall prevent undesired operation in any of the following cases:

11/13.8/33/34.5 kV Bus Fault (86B1, 86B2 and/or 86B3 operates).
11/13.8/33/34.5 kV Feeder Uncleared Fault (86-2 operates).
11/13.8/33/34.5 kV Bus Tie Lockout Operated (86BT-1/86BT-2).
Blocking of ABTS by Direct Intertrip Signal such that ABTS will not allow “Isolated Neutral System” for any other HV fault.

NOTE: ABTS shall also be blocked under double contingency conditions which may lead to isolated neutral system with arcing ground fault.

3.3.3 The ABTS shall ensure the following before giving any closing or tripping command:

The status of concerned MV circuit breakers (On or Off).
The position of concerned MV circuit breakers (in service or out of service).
The existence of voltage on the 11/13.8/33/34.5 kV buses.

3.3.4 No tripping command by ABTS to 11/13.8/33/34.5 kV incomers is required.

3.3.5 For the case where one of the normally paralleled transformers is lost, the tap positions of the other transformer shall be matched before transfer.

3.3.6 If the tripped transformer is the "Master" in the voltage control scheme, this function shall also be transferred to one of the transformers in service.

3.3.7 Independent of manual/automatic modes, after closing of the normally open bus tie breaker, it shall be tripped automatically if all the three (3) transformer incomers remain closed after a preset time delay of 45 seconds.

3.3.8 No automatic restoration of the scheme configuration is required. The normalization shall be manual only.

3.3.9 The transformers shall be on the same tap position before paralleling.

3.4 Automatic Switching Scheme For Four Transformers

The requirements shall be as indicated in the Project Technical Specifications/Scope of Work.

4. 11/13.8/33/34.5 kV AUTOMATIC CAPACITOR CONTROL SCHEME (ACCS)

4.1 AUTOMATIC CONTROL OF CAPACITOR BANK SYSTEM(ACCS) (FOR SUBSTATIONS WITH THREE TRANSFORMERS)

The automatic control of the capacitor banks shall be accomplished by the use of the Programmable Logic **C**ontrollers **(**PLC).The panel to house PLC and their associated devices of ACCS shall conform to 31-TMSS-01. The PLC shall be designed to be used in the substation environment. It shall have the watchdog system capability to monitor the PLC circuits and identify the defective circuit and give alarms. These alarms shall also be operated and monitored by SCADA. The programming procedures shall have following features.

4.1.1 The Human Machine Interface equipment shall be provided.

The PLC shall be flexible, extendable, if any additional input/output is needed by the Company Engineers. Therefore,the equipment needed to reprogram the microprocessor like servicing case, programming unit and video display unit shall be supplied by the Contractor in order to accomplish

the task at the field. It is the responsibility of the Contractor to train the Company personnel during the commissioning on the use and programming of the PLC. The PLC shall be equipped with its own DC/DC converter. It shall meet the standard of the IEC.

4.1.2 The PLC shall be completely devoted to Capacitor BankAutomatic Control The PLC shall not be used as annunciator.

However, annunciation for the PLC should be available for status, fault and operational progress by LED/LCD screens.

4.1.3 The Contractor shall be responsible for the automatic switching scheme in a fully operational order. Any material standard required shall be supplied. The proposed scheme shall be reviewed and approved by the company. The scheme shall be thoroughly tested during commissioning.

4.1.4 The ACCS shall be provided to perform the following functions minimum, but not limited to :

Automatic switching ON and OFF of the Capacitor Banks which shall be based on:

- Reactive Power measurement on the HV side of the power transformer.
- Voltage measured on the respective LV side of the bus as specified vide case -1 to case- VI below.
- ACCS shall function as per 43-TMSS-01

1. CASE-1

Bus coupler-1 (BC-1) and Bus Coupler-2 (BC-2) are in open position.

Capacitor Bank-1 (C-1) shall be switched ON when the reactive power of Power Transformer T-1 measured (HV side) is above the set value of the reactive power (MVAr-1) and the duration of this reactive power exceeds the set value of time (t-1). If the reactive power measured is below the set value MVAr-2 and the duration of the reactive power below this set value is above the set value of time (t-2), the Capacitor Bank (C-1) shall be switched OFF or remain in OFF condition.

Capacitor Bank -2 (C- 2) shall be switched ON when the reactive power of Power Transformer T-2 measured (HV side) is above the set value of the reactive power MVAr-1 and the duration of this reactive power exceeds the set value of time (t-1). If the reactive power measured is below the set value of reactive power MVAr-2 and the duration of this reactive power is above the set value of time (t-2), the Capacitor Bank - 2 (C-2) shall be switched OFF or remain in OFF condition.

Capacitor Bank-3 (C-3) shall be switched ON when the reactive power of the Power Transformer T-3 measured (HV side) is above the set value of the reactive power MVAr-1 and the duration of this reactive power exceeds the

set value of time (t-1). If the reactive power measured is below the set value of reactive power MVAr-2 and the duration of reactive power below this set value , is above this set value of time (t-2), the Capacitor Bank-3 (C-3) shall be switched OFF or remain in OFF condition.

2. CASE-II

Bus Coupler-1 (BC-1) in open position and Bus Coupler-2 (BC-2) in closed position

CapacitorBank -1 (C-1) shall be switched ON and OFF as per the same condition as specified under Case-1 for Capaciotor Bank-1 (C-1).

Capacitor Bank-2 (C-2) shall be switched ON when the total reactive power of the Power Transformers T-2 and T-3 measured (HV side) is above the set value of the reactive power (MVAr-3) and the duration of this reactive power exceeds the set value of time (t-3). If the total reactive power of Power Transformers T-2 and T-3 measured is below the set value of power (MVAr-4) and the duration of reactive power below the set value of reactive power, is above the set value of time (t-4) , the Capacitor Bank(C-2) shall be switched OFF or remain in OFF condition.

Capacitor Bank -3 (C-3) shall be switched ON when the total reactive power of Power Transformers T-2and T-3 measured (HV side) is above the set value of reactive power (MVAr-5) and the duration of this reactive power exceeds the set value of time (t-5). If the total reactive power of Transformers T-2 and T-3 measured is below the set value of reactive power (MVAr-6) and the duration of reactive power below the set value of reactive power, is above the set value of time (t-6), the Capacitor bank-3 (C-3) shall be switched OFF or remain in OFF condition.

3. CASE-III

Bus Coupler-1 (BC-1) is in Close position and the Bus Coupler-2 (BC-2) is in Open position

Capacitor-Bank 1 (C-1) shall be switched ON when the total reactive power of the Power Transformers T-1 and T-2 measured (HV side) is above the set value of reactive power (MVAr-3) and the duration of this reactive power exceeds the set value of time (t-3). If the total reactive power of Transformers T-1 and T-2 measured is below the set value of reactive power (MVAr-4) and the duration of reactive power below the set value of reactive power, is above the set value of time (t-4), the Capacitor Bank-1 (C-1)shall be switched OFF or remain in OFF condition.

Capacitor Bank-2 (C-2) shall be switched ON when the total reactive power of the Power Transformers T-1 and T-2 measured (HV side) is above the set value of reactive power (MVAr-5) and the duration of this reactive power exceeeds the set value of time (t-5). If the total power of Transformers T-1 and T-2 measured is below the set value of reactive power

(MVAr-6) and the duration of reactive power below the set value of reactive power, is above the set value of time (t-6), the Capacitor Bank-2(C-2) shall be switched OFF or remain in OFF condition.

Capacitor Bank -3 (C-3) shall be switched ON and OFF as per the same condition as specified under CASE-1 for CapacitorBank-3 (C-3).

4. CASE-IV

Bus Coupler-1 (BC-1) and Bus Coupler-2 (BC-2) are in close position

Capacitor Bank-1 (C-1) shall be switched ON when the total reactive power of Power Transformers T-1,T-2 & T-3 measured (HV side) is above the set value of the reactive power (MVAr-3) and the duration of this reactive power exceeds the set value of time (t-3). If the total reactive power of the Transformers T-1, T-2 & T-3 measured is below the set value of reactive power (MVAr-4) and the duration of reactive power below the set value of reactive power is above the set value of time (T-4), the Capacitor Bank -1 (C-1) shall be switched OFF or remain in OFF condition.

Capacitor Bank -2 (C-2) shall be switched ON when the total reactive power of Power Transformers T-1, T-2 & T-3 measured (HV side) is above the set value of reactive power (MVAr-5) and the duration of this reactive power exceeds the set value of time (t-5). If the total reactive power of Transformers T-1, T-2 & T-3 measured is below the set value of reactive power (MVAr-6) and the duration of reactive power below the set value of reactive power, is above the set value of time (t-6), the Capacitor Bank-2 (C-2) shall be switched OFF or remain in OFF condition.

Capacitor Bank-3 (C-3) shall be switched ON when total reactive power of Power Transformers T-1, T-2 & T-3 measured (HV side) is above the set value of reactive power (MVAr-7) and the duration of this reactive power exceeds the set value of time (t-7). If the total reactive power of Transformers T-1, T-2 & T-3 measured is below the set value of reactive power (MVAr-8) and the duration of reactive power below set value of reactive power is above the set value of time (t-8), the Capacitor Bank-3 (C-3) shall be switched OFF or remain in OFF condition.

5. CASE-V (OVERVOLTAGE CONTROL)

Automatic switching OFF of the Capacitor banks shall be performed based on the voltage on the respective bus; i.e when the bus voltage measures higher operating voltage (33 kV to 38 kV/13.8 kV to 16 kV). The Capacitor banks shall be switched OFF when the measured value of voltage is above the set voltage (V-1) and the duration exceeds the set value of time (t-9). This requirement shall be fulfilled even superceding the conditions of CASE-1 to IV.

6. CASE-VI (UNDER VOLTAGE CONTROL)

Automatic switching ON of the Capacitor banks shall be performed based on the voltage on the respective bus; i.e when the bus voltage measures lower operating voltage (28 kV to 33 kV/12 kV to 13.8 kV). The Capacitor banks shall be switched ON when the measured value of voltage is below the set voltage (V-2) and the duration exceeds the set value of time (t-10). This requirement shall be fulfilled even superseding the conditions of CASE-1 to IV above. ACCS shall also be provided with a facility to modify the setting and/or bypass/deactivate the settings, as required. In addition, ACCS shall prevent undesired switching in case of transient under voltage conditions.

4.1.5. ADDITIONAL CONTROL REQUIREMENT IF THE TOTAL CURRENT EXCEEDS THE CONTINUOUS CURRENT WITHSTAND CAPABILITY OF THE FILTER BANK

For all above cases, if the total current (including the harmonic current) exceeds the continuous current withstand capability of the Filter, the controller shall adopt the following strategy regardless of the reactive power flow in the Transformers.

a For the operating configuration of only one Filter connected to only one Transformer, the Filter should be switched OFF.

b For the operating configuration of only one Filter connected to two Transformers in parallel and if the measured voltage at 13.8 kV bus is below the set voltage (V-3) and duration exceeds the set value of time (t-11), the second connected Filter should be switched ON.

c For the operating configuration of only one Filter connected to two Transformers in parallel and if the measured voltage at 13.8 kV bus is above the set voltage (V-3) and duration exceeds the set value of time (t-11), this connected Filter should be switched OFF.

d For the operating configuration of two Filters connected to two Transformers in parallel, one of the two Filters should be switched OFF.

e For the operating configuration of three Filters connected to two Transformers in parellel, one of the three Filters should be switched OFF.

During these conditions, all the previous normal control logics should be superceded and suitable alarms and indications should be transmitted to Control center. Suitable provision for resetting the configuration/logic back to normal, only by manual resetting, should also be provided.

The minimum inputs to the ACCS shall consist of status of all 132 kV/33 kV & 132 kV/13.8 kV incomer (Transformer) CBs, bus tie CBs, each 33

kV/13.8 kV incomer voltage,,each 33 kV/13.8 kV bus voltage, each 33 kV/13.8 kV Capacitor Bank bus voltage, each Power Transformer's Var readings on the HV side, and status of lockout relays of incomer CBs/Capacitor CBs. Moreover, provision for overload protection shall be incorporated. When 33 kV/13.8 kV incomer (Transformer) CB is switched OFF , the associated Capacitor banks shall be switched OFF before 13.8 kV bus section CB is switched ON.

The CB of the Capacitor bank once switched OFF shall not switch ON within 10 minutes. Auto / Manual and Local / Remote selection shall be provided for the ACCS. In any case the 13.8 kV incomer CBs of Transformers T-1, T-2 & T-3 are switched ON, the associated Capacitor shall not switch ON within 10 minutes. Also two nos. Capacitor banks shall not be connected to a single Transformer.

The ACCS shall prevent undesired switching 'ON' operation of CBs of associated Capacitor banks in any one of the following cases:

i 33 kV/13.8 kV Bus fault (86 B1 or 86 B2 or 86B3) operates
ii 33 kV/13.8 kV Capacitor feeder unclear faults (main feeder CB or dedicated Capacitor feeder CB Lockout relay operates)
iii 33 kV/13.8 kV Bus Tie Lockout operates

Auto/Manual and Local/Remote selection shall be provided for the ACCS

4.1.6 RANGE OF SETTINGS

a. MVAr-1 Setting Ranges

MVAr-1 0 to 50 MVAr in steps of 1 MVAr
MVAr-2 0 to 50 MVAr in steps of 1 MVAr
MVAr-3 0 to 100 MVAr in steps of 1 MVAr
MVAr-4 0 to 100 MVAr in steps of 1 MVAr
MVAr-5 0 to 100 MVAr in steps of 1 MVAr
MVAr-6 0 to 100 MVAr in steps of 1 MVAr
MVAr-7 0 to 100 MVAr in steps of 1 MVAr
MVAr-8 0 to 100 MVAr in steps of 1 MVAr

b. Voltage Setting Ranges (for 33 kV)

V-1 33 kV to 38 kV in steps of 0.1 kV
V-2 28 kV to 33 kV in steps of 0.1 kV

c. Voltage Setting Ranges (for 13.8 kV)

V-1 13.8 kV to 16 kV in steps of 0.1 kV
V-2 12 kV to 13.8 kV in steps of 0.1 kV

d. Time Setting Ranges

t-1 0 to 30 minutes in steps of 1 minute
t-2 0 to 30 minutes in steps of 1 minute
t-3 0 to 30 minutes in steps of 1 minute
t-4 0 to 30 minutes in steps of 1 minute
t-5 0 to 30 minutes in steps of 1 minute
t-6 0 to 30 minutes in steps of 1 minute
t-7 0 to 30 minutes in steps of 1 minute
t-8 0 to 30 minutes in steps of 1 minute
t-9 0 to 120 seconds in steps of 0.1 second
t-10 0 to 5 seconds in steps of 0.1 second
t-11 0 to 5 seconds in steps of 0.1 second

For Substations with SAS, ACCS Panel shall be integrated and provided with dedicated control IED to carryout all ACCS functions.

4.2 13.8 kV AUTOMATIC CAPACITOR CONTROL SYSTEM (ACCS) FOR SUBSTATIONS WITH TWO TRANSFORMERS

The automatic control of Capcitor banks shall be accompanied by the use of Programmable Logic Controller (PLC). The panel to house PLC and other associated devices of ACCS shall conform to 31-TMSS-01. The PLC shall be designed to be used in the substation environment. It shall have the watchdog system capability to monitor the PLC circuit and identify the defective circuits and give alarms. These alarms shall be monitored by SCADA. It shall have an ON/OFF/TEST switch which will also be monitored by SCADA. The programming procedures shall be easily done without special knowledge of programming. The PLC shall be equipped with the following features:

4.2.1 The Human Machine Interface equipment shall be provided.

4.2.2 The PLC shall be flexible, extendible, if any additional input/output is needed. The PLC program shall be able to read in and modified, if needed in future by the COMPANY Engineers. Therefore, the equipment needed to reprogram the microprocessor such as the serving case, programming unit and video display unit shall be supplied by the CONTRACTOR in order to accomplish this task in the field. It is the responsibility of the CONTRACTOR to train COMPANY personnel during commissioning on the use and programming of the PLC. The PLC shall be equipped with its own DC/DC convertor. It shall meet the standard of the IEC.

4.2.3 The PLC shall be completely devoted to Capacitor Bank Automatic control.The PLC shall not be used as annunciator.

However, annunciation for the PLC should be available for status, fault and operational progress by LED/LCD screens.

4.2.4 The CONTRACTOR shall be responsible for the Automatic switching scheme in a fully operational order. Any material required shall be supplied.

The proposed scheme shall be reviewed and approved by the COMPANY. The scheme shall thoroughly be tested during commissioning.

4.2.5 The ACCS shall be provided to perform the following functions minimum,but not limited to:

Automatic switching 'ON' and 'OFF' of the Capacitor bank shall be based on;

i) Reactive Power measurement on the HV side of the 132/13.8 kV Transformers.
ii) Voltage measured on the respective 13.8 kV buses as specified vide cases I-IV below.

1. CASE-I

Bus Sectionaliser (BS) in open position

Capacitor Bank -1 (C-1) shall be switched ON when the reactive power of Power Transformer T-1 measured HV side is above the set value of reactive power (MVAr-1) and the duration of this reactive power exceeds the set value of time(t_1). If the reactive power measured is below the set value of reactive power (MVAr-2) and the duration of the reactive power below this set value exceeds the set value of time (t_2), the Capacitor Bank -1 (C-1) shall be switchd OFF or remain in OFF position.

Capacitor Bank -2 (C-2) shall be switched ON when the reactive power of the Power Transformer T-2 measured (HV side) is above the set value (MVAr-1) and the duration of this reactive power exceeds the set value of time (t_1). If the reactive power measured is below the measured value of reactive power (MVAr-2) and the duration of this reactive power below the set value exceeds the set value of time (t_2), then the Capacitor Bank-2 (C-2) shall be switched OFF or remain in OFF position.

2. CASE-II

Bus Sectionalizer (BS) is in closed position

Capacitor Bank -1 (C-1) shall be switched ON when the total reactive power of Power Transformers T-1 & T-2 measured (HV side) is above the set value of reactive power (MVAr-3) and the duration of this reactive power exceeds the set value of time (t_3). If the total reactive power of the Power Transformers T-1 & T-2 measured is below the set value of reactive power (MVAr-4) and the duration of reactive power below the set value of reactive power exceeds the set value of time (t_4), the Capacitor Bank-1 (C-1) shall be switched OFF or remain in OFF position.

CapacitorBank-2 (C-2) shall be switched ON when the total reactive power of the Power Transformers T-1 & T-2 measured (HV side) is above the set value of reactive power (MVAr-5) and the duration of this reactive power exceeds the set value of time (t_5). If the total reactive power of Power Transformers T-1 & T-2

measured (HV side) is below the set value of reactive power (MVAr-6) and the duration of reacitve power below the set value of reactive power exceeds the set value of time (t_6), then the Capacitor Bank-2 (C-2) shall be switched OFF or remain in OFF position.

3. CASE-III (OVER VOLTAGE CONTROL)

Automatic switching OFF of the Capacitor banks shall be performed based on the voltage measured on the respective bus; i.e when the bus voltage measures higher operating voltage (13.8 kV – 16 kV). The Capacitor banks shall be switched OFF when the measured value of voltage is above the set value of voltage (V-1) and duration exceeds the set value of time (t-9). This requirement shall be fulfilled even superseding the conditions of CASES I & II above.

4. CASE-IV (UNDER VOLTAGE CONTROL)

Automatic switching ON of the Capacitor banks shall be performed based on the voltage measured on the respective bus; ie when the bus voltage measures lower operating voltage (13.8 kV – 12 kV). The Capacitor banks shall be switched ON when the measured value of voltage is below the set value of voltage (V-2) & duration of time exceeeds the set value of time (t_{10}). This requirement shall be fulfilled even when superseding the conditions of CASES I & II above.

ACCS shall also be provided with a facility to modify the setting and/or bypass/deactivate the settings, as required. The minimum inputs to the ACCS shall consist status of all 132 kV and 13.8 kV incomer (Transformer) CB(s), bus tie CBs, each 13.8 kV incomer voltage, each 13.8 kV bus voltage, each Capacitor bank bus voltage, each Power Transformer's VAR readings on the HV side, and status of lockout relays of incomer CB(s)/Capacitor CB(s). Provision for unbalanced protection shall also be incorporated. ACCS shall prevent undesired switching in case of transient under voltage conditions.

When 13.8 kV incomer (Transformer) CB is switched OFF, the associated Capacitor banks shall be switched OFF before 13.8 kV bus section CB is switched ON .

The CB of the Capacitor bank once switched OFF shall not switch ON within 10 minutes. Auto / Manual and Local / Remote selection shall be provided for the ACCS. In any case the 13.8 kV incomer breakers of Transformers T-1 or T-2 shall be switched ON , the associated Capacitor shall not switch ON within 10 minutes.

The ACCS shall prevent undesired switching ON operation of CBs of associated Capacitor banks in any one of the following cases.

i. 13.8 kV Bus fault
ii 13.8 kV Capacitor feeder unclear faults (main feeder CB or dedicated Capacitor feeder CB lockout relay operates)
iii 13.8 kV Bus tie lockout operates

Auto/Manual and Local/Remote selection shall be provided for the ACCS

4.2.6 RANGE OF SETTINGS

a. MVAr Setting Ranges

MVAr-1	0 to 50 MVAr	in steps of	1 MVAr
MVAr-2	0 to 50 MVAr	in steps of	1 MVAr
MVAr-3	0 to 100 MVAr	in steps of	1 MVAr
MVAr-4	0 to 100 MVAr	in steps of	1 MVAr
MVAr-5	0 to 100 MVAr	in steps of	1 MVAr
MVAr-6	0 to 100 MVAr	in steps of	1 MVAr

b. Voltage Setting Ranges

V-1	13.8 to 16 kV	in steps of	0.1 kV
V-2	12 to 13.8 kV	in steps of	0.1 kV

c. Time Setting Ranges

t-1	0 to 30 minutes	in steps of	1 minute
t-2	0 to 30 minutes	in steps of	1 minute
t-3	0 to 30 minutes	in steps of	1 minute
t-4	0 to 30 minutes	in steps of	1 minute
t-5	0 to 30 minutes	in steps of	1 minute
t-6	0 to 30 minutes	in steps of	1 minute
t-7	0 to 120 Seconds	in steps of	1 second
t-8	0 to 5 Seconds	in steps of	0.1 second

For Substations with SAS, ACCS Panel shall be integrated and provided with dedicated control IED to carryout all ACCS functions.

5.0 BIBLIOGRAPHY

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